

Rishabh Bharatbhai Parmar

AI/ML Engineer

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Professional Summary

Performance-driven AI/ML Engineer with hands-on expertise in **Computer Vision**, **Edge AI**, and applied **Generative AI**. Experienced in designing, deploying, and accelerating real-time deep learning pipelines for **classification**, **object detection**, **segmentation**, **OCR**, and **tracking** across **NVIDIA Jetson**, **NPU**s, and **embedded SBC**s. Skilled in hardware-aware optimization using quantization, pruning, TensorRT, and ONNX, achieving up to **30% faster inference** throughput and **50% lower memory** consumption. Experienced in multi-protocol camera ingestion (RTSP, MIPI, CSI, USB) and backend API integration using FastAPI. Actively building applications leveraging **LLMs** and **RAG architectures** for real-world use cases.

Work Experience

Junior AI/ML Engineer

Webocult Technologies

Sep 2025 – Aug 2026

Ahmedabad, Gujarat, India

- Designed, trained, and deployed end-to-end **Computer Vision inference pipelines** for real-time **Edge AI** applications across diverse embedded hardware platforms.
- Built and optimized multi-task Computer Vision pipelines for classification, detection, segmentation, and OCR using YOLO and PaddleOCR, achieving real-time inference across **CPU, GPU, and NPU runtimes**.
- Engineered and benchmarked hardware-aware model optimization and quantization pipelines, improving inference throughput by 30% and reducing memory consumption by up to 50% on resource-constrained edge devices.
- Integrated multi-protocol video input streams including USB, IP (RTSP), CSI, and MIPI cameras, ensuring resilient video ingestion and fault-tolerant stream processing.
- Ported and optimized vision models across **NXP i.MX 8M Plus**, **NVIDIA Jetson**, **Rockchip RKNN**, **Axelera AI**, **MediaTek Arbor SBC**, **Amobile G700**, and **Raspberry Pi IMX500**, adapting inference pipelines to platform-specific hardware constraints.
- Developed and fine-tuned custom vision models (mobile phone detection, item/skittle detection, and container tracking) integrated with CPU-accelerated OCR engines.
- Automated Linux edge deployment workflows, environment setup, and pipeline monitoring using robust Bash/Shell scripts.
- Engineered showcase Edge AI prototypes exhibited at global technology expos including **CES**, **Embedded World**, and **EuroShop**.
- Conducted **on-site client deployments**, live validation, hardware calibration, and real-time troubleshooting under real-world operational constraints.

Trainee AI/ML Engineer

Webocult Technologies

Jan 2025 – Aug 2025

Ahmedabad, Gujarat, India

- Worked with OpenCV, NLP, dataset annotation, and AI inference pipeline development on **DeepX** and **Deeper-i** NPUs.
- Developed and deployed an end-to-end Edge AI car parking solution for **Japan IT Week** in collaboration with Deeper-i, enabling real-time parking space detection and monitoring.
- Built FastAPI-based REST APIs for AI model inference and integration with application backends.

AI & ML Intern

Edunet Foundation

May 2024 – Jun 2024

Anand, Gujarat, India

- Built a Driver Behavior Monitoring System utilizing computer vision and neural networks for real-time driver state and facial landmark analysis.
- Developed image pre-processing pipelines and automated data scraping scripts using Python and OpenCV for efficient dataset collection and processing.

Key Projects

LocalDoc AI – Local AI-Powered Document QA & Multi-Turn RAG System

Generative AI / RAG

Technologies: Python, LangChain, ChromaDB, Hugging Face Transformers, Sentence-Transformers, FastAPI, React.js

- Built an end-to-end **privacy-first local RAG system** with semantic chunking and persistent vector retrieval using **ChromaDB**.
- Developed **threshold-based adaptive query routing** that evaluates semantic similarity scores to distinguish document-grounded queries from general conversational prompts, reducing irrelevant retrieval and hallucinations.
- Architected **dual-layer rolling conversation memory** (recent context + neural summary) optimized for the **360M-parameter LLM**, with Hugging Face chat templates, FastAPI backend, and a responsive React/Tailwind interface with source citations and retrieval-score inspection.

VirtuDrum – Real-Time AI-Driven Virtual Drumming System

Desktop Application

Technologies: Python, OpenCV, MediaPipe, Qt/PyQt, Flask, SQLite

- Designed an interactive virtual drumming application leveraging real-time 3D hand landmark tracking and gesture recognition via MediaPipe and OpenCV.
- Built an event-driven modular architecture featuring low-latency Air Drum triggering, rhythm gameplay modes, and cross-platform desktop execution.
- Implemented dual-mode local/cloud score synchronization and user profiling using a lightweight Flask REST backend and SQLite database.

- Developed an intelligent in-cabin safety system analyzing facial landmarks and eye aspect ratios in real time to detect drowsiness, distraction, and fatigue.
- Integrated YOLOv8 object detection for real-time mobile phone usage detection and negative emotion classification, triggering immediate safety alerts.
- Optimized frame processing and pipeline throughput to sustain reliable real-time FPS on constrained edge compute units.

Achievements & Honors

- **Winner – Smart India Hackathon (SIH) 2023:** Secured **1st place** nationwide for our problem statement at **Smart India Hackathon (SIH) 2023**, earning an **INR 1,00,000** cash prize.
- **Pre-Qualified – Smart India Hackathon (SIH) 2022:** Shortlisted at the institutional level among hundreds of participating teams.
- **NPTEL Star (Enthusiast & Believer) – 2022 & 2023:** Recognized as an NPTEL Star by clearing **12 consecutive technical certification exams** in a single attempt.
- **Global Contributor – Hacktoberfest 2022:** Authored and merged 5 pull requests across 2 open-source repositories; ranked in the top 40,000 global contributors.

Education

Birla Vishvakarma Mahavidyalaya (BVM) <i>Bachelor of Technology in Information Technology</i> • Academic Record: CPI: 8.55 / 10 (80.5%)	2021 – 2025 <i>Anand, Gujarat, India</i>
The Aditya Birla Public School <i>Higher Secondary Certificate (HSC) – Class XII (CBSE)</i> • Academic Record: Percentage: 90.4%	2020 – 2021 <i>Bharuch, Gujarat, India</i>
The Aditya Birla Public School <i>Secondary School Certificate (SSC) – Class X (CBSE)</i> • Academic Record: Percentage: 87.0%	2018 – 2019 <i>Bharuch, Gujarat, India</i>

Technical Skills

- **Languages:** Python, C++, SQL, Bash / Shell Scripting.
- **Fundamentals:** Computer Vision, Multi-Object Tracking, Machine Learning, Deep Learning, Natural Language Processing (NLP), Transfer Learning, Model Quantization.
- **AI/ML Frameworks:** PyTorch, OpenCV, TensorRT, ONNX Runtime, YOLO Models, PaddleOCR, MediaPipe, Ultralytics.
- **Generative AI & LLMs:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Vector Databases (ChromaDB), Text Embeddings, Multimodal Pipelines.
- **Edge AI & Embedded Platforms:** NVIDIA Jetson (Orin Nano, AGX Orin, Thor), NXP i.MX 8M Plus, Rockchip RKNN, Axelera AI, Sony IMX500, DeepX NPU, Deeper-i NPU, MediaTek Arbor SBC, Amobile G700, Raspberry Pi.
- **Backend:** FastAPI, Flask, RESTful APIs, MongoDB, SQLite, Socket.
- **Tools:** Linux (Ubuntu), Git, GitHub, Docker, Postman, VS Code, Antigravity, TagSpaces (S3).

Leadership & Volunteer Experience

- **Chairperson** | *GeeksforGeeks (GFG) Student Chapter, BVM* Sep 2023 – May 2024
 - Led a core student team to organize technical bootcamps, coding contests, and developer workshops impacting 500+ student participants.
- **Web Developer** | *CodeChef BVM Chapter* Sep 2022 – Apr 2023
 - Contributed to frontend web development and platform maintenance for campus-wide competitive programming initiatives.
- **Technical Core Member** | *Google Developer Student Clubs (GDSC) – BVM* Sep 2022 – Apr 2023
 - Assisted in orchestrating open-source workshops, hackathons, and study jams on modern software development and cloud technologies.